

9.6.3

EE25BTECH11054 - S.Harsha Vardhan

Question 9.6.3 Find the area of the circle $x^2 + y^2 = 16$ exterior to the parabola $y^2 = 6x$.

Solution: The total area of the circle $x^2 + y^2 = 16$ is $A_{circle} = \pi r^2 = 16\pi$. First, we find the points of intersection of the two curves. The conic parameters for the two curves are: For the circle $x^2 + y^2 - 16 = 0$:

$$\mathbf{V}_1 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad \mathbf{u}_1 = \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \quad f_1 = -16. \quad (1)$$

For the parabola $y^2 - 6x = 0$:

$$\mathbf{V}_2 = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}, \quad \mathbf{u}_2 = \begin{pmatrix} -3 \\ 0 \end{pmatrix}, \quad f_2 = 0. \quad (2)$$

The intersection of the two conics is given by the equation:

$$\mathbf{x}^T (\mathbf{V}_1 + \mu \mathbf{V}_2) \mathbf{x} + 2(\mathbf{u}_1 + \mu \mathbf{u}_2)^T \mathbf{x} + (f_1 + \mu f_2) = 0 \quad (3)$$

For a degenerate conic, the determinant of the quadratic part's matrix must be zero.

$$\det(\mathbf{V}_1 + \mu \mathbf{V}_2) = 0 \quad (4)$$

$$\det \left(\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + \mu \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \right) = \begin{vmatrix} 1 & 0 \\ 0 & 1 + \mu \end{vmatrix} = 1(1 + \mu) = 0 \quad (5)$$

$$\implies \mu = -1 \quad (6)$$

Substituting $\mu = -1$ into equation (3):

$$\mathbf{x}^T (\mathbf{V}_1 - \mathbf{V}_2) \mathbf{x} + 2(\mathbf{u}_1 - \mathbf{u}_2)^T \mathbf{x} + (f_1 - f_2) = 0 \quad (7)$$

$$\mathbf{x}^T \left(\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} - \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \right) \mathbf{x} + 2 \left(\begin{pmatrix} 0 \\ 0 \end{pmatrix} - \begin{pmatrix} -3 \\ 0 \end{pmatrix} \right)^T \mathbf{x} + (-16 - 0) = 0 \quad (8)$$

$$\mathbf{x}^T \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \mathbf{x} + 2 \begin{pmatrix} 3 \\ 0 \end{pmatrix}^T \mathbf{x} - 16 = 0 \quad (9)$$

Letting $\mathbf{x} = \begin{pmatrix} x \\ y \end{pmatrix}$, the equation becomes:

$$\begin{pmatrix} x & y \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} + 2 \begin{pmatrix} 3 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} - 16 = 0 \quad (10)$$

$$\implies x^2 + 6x - 16 = 0 \quad (11)$$

Solving for x : $(x + 8)(x - 2) = 0$, which gives $x = 2$ or $x = -8$. Substituting these values into the parabola's equation $y^2 = 6x$: If $x = 2$, $y^2 = 12 \implies y = \pm \sqrt{12} = \pm 2\sqrt{3}$. If

$x = -8$, $y^2 = -48$, which has no real solution for y . Hence, the points of intersection are:

$$\mathbf{a}_1 = \begin{pmatrix} 2 \\ 2\sqrt{3} \end{pmatrix}, \quad \mathbf{a}_2 = \begin{pmatrix} 2 \\ -2\sqrt{3} \end{pmatrix} \quad (12)$$

$$A_{\text{interior}} = 2 \left[\int_0^2 \sqrt{6x} dx + \int_2^4 \sqrt{16-x^2} dx \right] \quad (13)$$

First integral:

$$\int_0^2 \sqrt{6x} dx = \sqrt{6} \left[\frac{2}{3} x^{3/2} \right]_0^2 = \frac{2\sqrt{6}}{3} (2\sqrt{2}) = \frac{4\sqrt{12}}{3} = \frac{8\sqrt{3}}{3} \quad (14)$$

Second integral (using $\int \sqrt{a^2 - x^2} dx = \frac{x}{2} \sqrt{a^2 - x^2} + \frac{a^2}{2} \sin^{-1} \frac{x}{a}$):

$$\begin{aligned} \int_2^4 \sqrt{16-x^2} dx &= \left[\frac{x}{2} \sqrt{16-x^2} + \frac{16}{2} \sin^{-1} \left(\frac{x}{4} \right) \right]_2^4 \\ &= (0 + 8 \sin^{-1}(1)) - \left(\frac{2}{2} \sqrt{16-4} + 8 \sin^{-1} \left(\frac{1}{2} \right) \right) \\ &= 8 \left(\frac{\pi}{2} \right) - \left(\sqrt{12} + 8 \left(\frac{\pi}{6} \right) \right) = 4\pi - 2\sqrt{3} - \frac{4\pi}{3} = \frac{8\pi}{3} - 2\sqrt{3} \end{aligned} \quad (15)$$

Adding the results for A_{interior} :

$$A_{\text{interior}} = 2 \left[\frac{8\sqrt{3}}{3} + \frac{8\pi}{3} - 2\sqrt{3} \right] = 2 \left[\frac{2\sqrt{3}}{3} + \frac{8\pi}{3} \right] = \frac{4\sqrt{3}}{3} + \frac{16\pi}{3} \quad (16)$$

The required area exterior to the parabola is $A_{\text{exterior}} = A_{\text{circle}} - A_{\text{interior}}$.

$$A_{\text{exterior}} = 16\pi - \left(\frac{4\sqrt{3}}{3} + \frac{16\pi}{3} \right) = \frac{48\pi - 16\pi}{3} - \frac{4\sqrt{3}}{3} = \frac{32\pi - 4\sqrt{3}}{3} \quad (17)$$

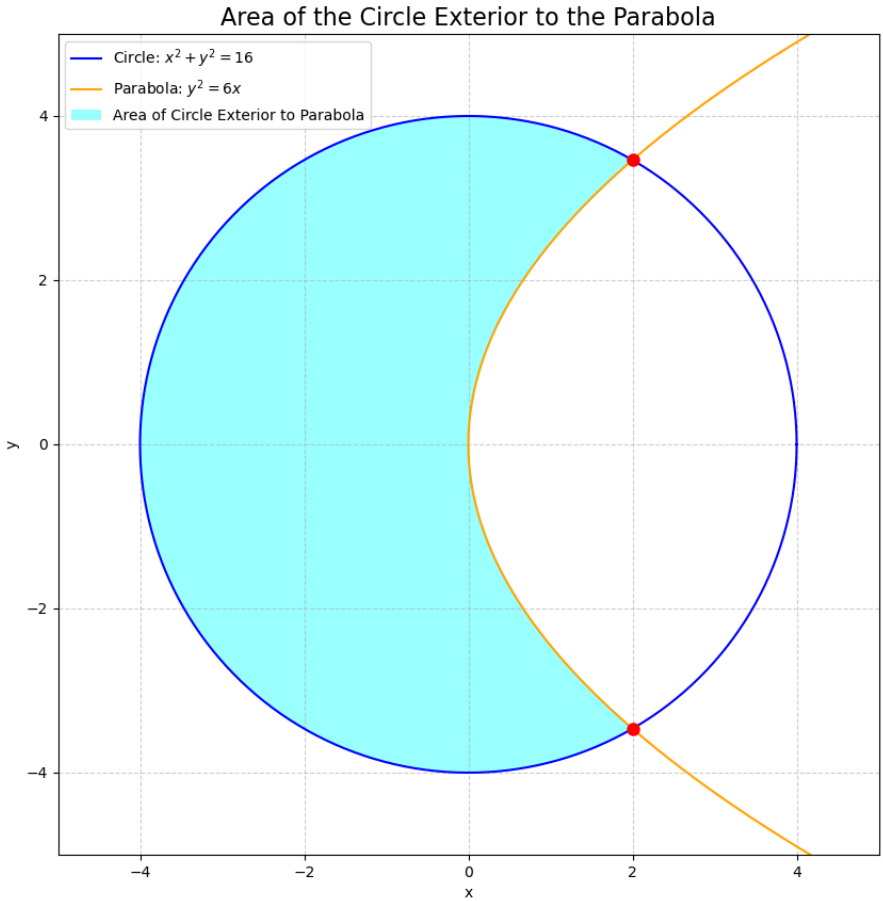


Fig. 0.1